Component Rebuild Center (CRC) TAT Measurement Application (CTM)Development

Gainwell Commosales Private Limited (GCPL)

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Ver 1.1

From

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About OPA Technology Solutions LLP

1.1 About US

OPA Technology Solutions (hereinafter called "OPA") has been established on Jan'2017 based out of Kolkata, India to provide the Software Product and IT solutions to the customer across the globe. It has been established by the Ex-Cognizant leaders and their vision is to solving problems of the customer rather than giving IT solutions only. While providing the solution to the business problems, OPA takes the ownership of implementing the solution as well as running the application to align the ongoing business challenges of our customers. Following are some of the services / solutions provided by OPA:

- Product and services in eCommerce and Master Data Management Space
- IT Services like Website Development, Maintenance & Hosting
- Analytics
- Mobile App Development
- Consulting in Healthcare and eCommerce space

1.2 Background and Objective of CRC TAT Measurement Project

GCPL commissions caterpillar assets in their customer place and they have CRC department who are responsible to rebuild the asset or component in the Asansol center. In the process of rebuilding the assets / components, CRC department follow the process and want to measure the turnaround time (TAT) of the activities based on some KPIs. The purpose of developing this solution is to cater to following objectives:

- Measure the TAT vs benchmark based on the KPIs
- Create a collaboration and notification platform to manage the asset / component rebuild
- Manage the rebuild process activities efficiently with optimum manual intervention
- Status tracking of the rebuild process activities centrally
- Manage preventive maintenance during warranty period
- Increase in operation efficiencies

OPA is invited to deliver a complete custom made **Responsive Web Application** solution for CRC TAT measurement (CTM) project.

Scope of Work

The scope of work includes design, development and implementation, Testing and Quality Assurance of a **Responsive Web Application development for** CRC TAT measurement

project of the assets commissioned in their customer location. The detail scopes are mentioned below.

GCPL's CRC TAT measurement project will have following high level functionalities which will help to fulfill the objectives of this project:

- **Admin Interface**
  - Admin Dashboard to view following:
    - For all work order, Section wise TAT vs. benchmark. Typically sections are Receive, Engine & Transmission, Quality, Parts Requisition and Dispatch
    - Work order wise section wise TAT vs. benchmark
    - Work order wise & section wise defaulter list meaning defaulters who are deviating from benchmark and reason for defaulting
  - CTM Website Content Management
  - Master Data Management like maintenance of work status, branch, section wise checklist template etc.
  - Configure workflow for notification
- **Service Branch Office (Service Engineers and Manager)**
  - Branch manager dashboard on preventive maintenance during warranty period
  - Create service request through service request form
  - Upload pre-history data of asset or component
  - Configure / Upload warranty preventive maintenance schedule
  - Enter preventive maintenance checklist
- **CRC Branch**
  - CRC Head Dashboard which will be similar to Admin dashboard
  - Create work order for a customer
  - CTM System will create the rebuild plan automatically based on the benchmark data configured in the system
  - Create and update quotation for the rebuild
  - Generate and send quotation to branch
  - Upload PO or confirmation of the work
- **CRC Receiving Section**
  - Create receive checklist along with photo
  - Notification using workflow to dismantling department
- **CRC Engine & Transmission Section**
  - **Dismantling**
    - Gets notification through system from receiving department
    - View all the notifications in the CTM system

- Select relevant work order and create dismantling checklist and update the status
- Select relevant work order and create inspection checklist and update the status

- **Assembly Section**
  - Gets notification of material requisition status
  - View all the notifications in the CTM system
  - Select relevant work order and create assembly checklist
  - System send notification for BIS to Quality department
  - Create pre invoice checklist
  - CTM System sends AIS notification to quality

- **CRC Power trains – hydraulic Components Section**
  - Gets notification from CTM system on the schedule for hydraulic components
  - View all the notifications in the system
  - Select relevant work order and create hydraulic component checklist and update the status including the reason for delay (if any)
- **CRC Machine Repair & Overhaul (MRO) Section**
  - Gets notification from CTM system on the schedule for MRO
  - View all the notifications in the system
  - Select relevant work order and create MRO checklist and update the status including the reason for delay (if any)
- **CRC Quality Section**
  - Gets the BIS notification from the CTM system
  - View all the notifications and details in the CTM system
  - Create BIS checklist in the CTM system
  - Generate test certificate from the CTM system
  - Gets AIS notification from CTM system
  - Create pre dispatch checklist
- **CRC Parts Requisition Section**
  - Gets the notification for parts requisition
  - View the parts notification details and update the status including the reason for delay (if any)
  - Parts department checks the availability in SAP and this is not going to be part of CRC portal
  - Create receipt of challan
- **CRC Dispatch Section**
  - Gets the notification of AIS
  - Create dispatch checklist along with photo

- **Integration Needs**
  - Offline integration with SAP asset / component data
  - Mail system integration for workflow notification

Scope Assumptions:

Following are the assumptions of the scope of work as mentioned above:

- **Any changes to the scope mentioned above will be treated as change** request and OPA will assess the impact of change and will submit the change effort and price separately
- Apart from SAP there will not be any more integration needed for this project
- CTM application will have responsive so that it can be accessed from mobile devices
- Mobile App development is not within the scope of this proposal
- Hosting charges of the CTM application is not considered within this scope
- User Acceptance Testing (UAT) will be conducted by GCPL rather than any other 3rd party like GCPL's customers

Technology, Solution & Timelines

Solutions:

OPA proposes to have Web Application which will have common middleware to interact with the database. The SAP will provide data to be pushed to the web middle tier interface. CRON jobs will be configured to fetch data from SAP to Web Middle Tier and vice versa.

Technology:

OPA proposes to adopt following technology stack to optimize the technology cost of ownership and development timeline:

- Framework: ionic 3 / ionic 4 for Mobile App and Laravel for Web middle tier
- Front End: Java Script, HTML5, CSS3
- Distributed By: Apache Cordova
- Database: MySQL Server
- Web Server: Apache

Timelines:

OPA will follow Agile development methodology considering the urgency of GCPL and thereby reducing the time for modules to be available to GCPL and ensuring enough time for modules / on-the-fly integration testing. OPA shall complete the design and development of the solution within **10-12** calendar weeks from the date of releasing the purchase order.

Weeks	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Activities												
Requirement Analysis												
Architecture & Design												
Prototype												
Runnable Demo for Inaugaration												
Development												
System & Integration Testing												
UAT												
Production Roll-out												

Following are the deliverables from OPA:

Sl. No.	Deliverables
1.	Requirement Document / High Level Design Document
2.	Prototype screens

Sl. No.	Deliverables
3.	Code Base
4.	User Manual

Adherence to timelines is closely linked to timely reviews and feedback from company stakeholders. It is expected that the review cycle for each deliverable will not exceed 4 calendar days. In absence of any feedback within 4 calendar days, the deliverable will be deemed to be accepted and approved by the stakeholders.

Any alterations in the initially agreed requirement and specifications will / might incur additional time and charges upon agreement by both the parties. Additionally any further enhancement or maintenance post-delivery will not be covered under this proposal and would thereby need to be framed as and when required.

Commercials

Price:

OPA will charge a consolidated service fee of **INR 2,84,000 (Indian Rupees Two Lakh Eighty Four Thousand Only)** for designing and developing CTM Application. The above service fee, provided by OPA is **exclusive of applicable service taxes** and all other applicable levies as defined by the Government of India, which will be borne by GCPL.

Payment Schedule:

The payment schedule of service fees to OPA for the CTM application development is shown below:

Sl. No.	Milestone	Percentage	Amount
1.	Advance Payment	20%	Rs. 56,800
2.	Delivery of Demo	20%	Rs. 56,800
3.	UAT	20%	Rs. 1,42,000
4.	After Warranty	10%	Rs. 28,400
		Total	**Rs. 2,84,000**

Pricing Assumptions:

- GCPL needs to provide networked / server infrastructure and access to their internal network to OPA for ease of development and integration.
- SAP Interfaces of GCPL should be ready for consumption by the CTM Application.

Warranty:

Post-UAT, the warranty support will be provided by OPA, **free of charge, for 3 Months (Three Month).**

General Terms and Conditions:

Mode of Payment

All payments are to be made by wire transfer to OPA. The account details will be communicated in due course. Alternatively payments to OPA can be made through cheque in favor of "OPA Technology Solutions LLP" payable at Kolkata.

OPA charges mentioned here-in are exclusive of all applicable taxes.

If Income Tax is deducted at source as per the existing Tax laws for any payment(s), **GCPL will provide corresponding Income Tax Deduction certificate as soon as they are available (no later than next quarter of the financial year in which the deduction has been made).**

Language of Communication

The official language of communication will be English for this project.

Lead Time

OPA as a whole will start working on the assignment within 5 days of receiving the PO from GCPL.

Non-employment

GCPL will neither offer employment nor employ, directly or otherwise, any OPA employee, associated with the assignment, during the period between the date of the proposal and one year from the completion of the assignment. Similarly OPA will neither offer employment nor employ, directly or otherwise, any GCPL employee, associated with the assignment, during the period between the date of the proposal and one year from the completion of the assignment.

Force Majeure

If the performance, as specified in this proposal is prevented, restricted, delayed or interfered by reason of:

Fire, explosion, cyclone, floods;
War, revolution, acts of public enemies, blockage of embargo;
Any law, order proclamation, ordinance, demand or requirements of any Government or authority or representative of any such Government, including restrictive trade practices or regulations;
Strikes, shutdowns or labor disputes which are not instigated for the purpose of avoiding obligations herein;
Or
Any other circumstances beyond the control of the party affected;

then notwithstanding anything here before contained, the party affected shall be excused from its performance to the extent, such performance relates to the prevention, restriction, delay or interference, and provided the party so affected uses its best efforts to remove such cause of non-performance, and when removed the party shall continue performance with the utmost dispatch.

Arbitration

Any dispute or difference of opinion between OPA and GCPL during the course of the proposed assignment will be referred for arbitration to a Board of Arbitration. This Board will be constituted prior to the commencement of the arbitration and will consist of two representative arbitrators and an umpire, appointed by the arbitrators. Arbitration will be carried out by Bengal National Chamber of Commerce Kolkata.

Termination

OPA and GCPL will have the right to terminate the agreement resulting from the proposal by giving 30 days' notice to the other party in case of breach of trust or breach of contract or non-compliance with the stipulations of the proposal. The termination notice will be held

valid only if it is preceded by a corresponding non-compliance notice issued at least 30 (thirty) days prior to the date of the termination notice and if non-compliance has continued up to the date of the termination notice.

If OPA issues the termination notice, GCPL will have to clear all payments due to OPA as of the date of the termination notice. These payments will cover the proportion of work completed in the on-going month until termination.

If GCPL issues the termination notice, they will pay OPA all amounts due as of the date of the termination notice as well as the payment that would accrue on completion of work for the month encompassing the termination date.— **Purely subject to reason of termination**

Cost for disengagement during termination will be borne by the party who will be issuing notice for termination.

Validity Period

This proposal and the commercials are valid for a period of 90 days from the date of submission. Beyond this period, the commercials and other clauses have to be re-validated by OPA.